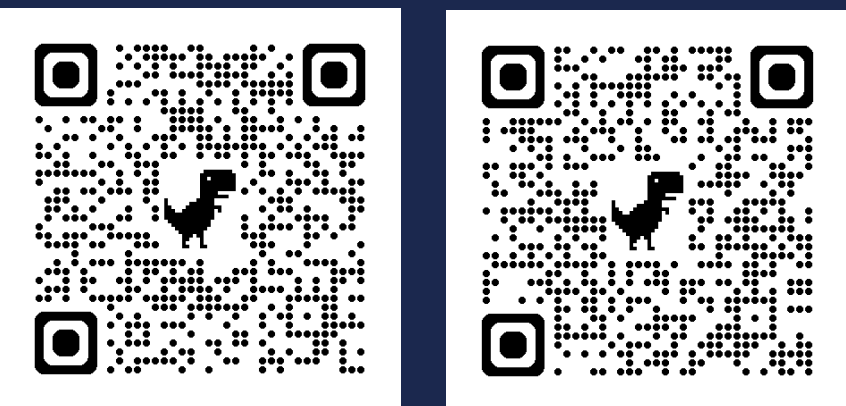




BAYESIAN ACTIVE LEARNING WITH GAUSSIAN PROCESSES GUIDED BY LLM RELEVANCE SCORING FOR DENSE PASSAGE RETRIEVAL



Paper Code

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ACL 2026
SAN DIEGO JULY 2-7

RESEARCH QUESTION

LLMs excel at relevance scoring but are too expensive for large corpora.
→ Passage retrieval as a **budget-constrained global optimization**.

Efficient search in **vast spaces** with **limited LLM inferences**.

LIMITATIONS OF RERANKING PARADIGM

LLMs passively rely on first-stage candidate set.

Recall Bottleneck

Retrieval is limited by the first-stage retriever's local neighborhood.

→ If missed early, the LLM never sees it.

Isolated Signals

Scoring a passage provides no information about its neighbors.

→ Evaluating one tells nothing about the rest.

💡 Solution: **Gaussian Process(GP)-based Active Learning!**

!! Takeaways

	Reranking	BAGEL
Search Strategy	✗ Passive (Fixed-set rescoring)	✓ Active (via Bayesian optimization)
Search Range	✗ Local (Neighborhood near query)	✓ Global (Entire corpus)
Signal Efficiency	✗ Independent (Per-passages)	✓ Propagated (via Gaussian Process)

WHY GP-BASED ACTIVE LEARNING?

- Kernel-based Relevance Propagation**
→ **Interpolates signals** across the embedding space to model **global multimodal relevance**.
 - Uncertainty for Active Learning**
→ Using **probabilistic posteriors** to target promising and unexplored areas to **maximize LLM efficiency**.
- 🧠 **BAGEL propagates LLM relevance signals** across the embedding space to guide **global exploration**.

BAGEL: BAYESIAN ACTIVE LEARNING WITH GAUSSIAN PROCESS GUIDED BY LLM RELEVANCE SCORING

🧠 **BAGEL produces a full corpus ranking under a limited LLM budget by interpolating sparse signals.**

1. Query-Specific GP

- A surrogate of the LLM relevance function
- The kernel propagates signals to the entire corpus.
- Kernel matters (Stationary kernels capture multi modal distributions.).

1. Query-Specific Gaussian Process
Query, Observed Passages $\mathcal{D}_q$
Query-passage relevance function $f_q^{LLM} \sim \mathcal{GP}(0, k_q(\cdot, \cdot))$

2-(a) Passage Selection via Warm-start Initialization

- Initialize with Query + Top-M passages from dense retrieval
- Provides high-quality signals to mitigate the cold-start problem

2-(b) Passage Selection via Active Learning

- Iteratively select passages with an acquisition function
- Uncertainty is the key!
- Balance exploitation (relevance) + exploration (uncertainty)

EXPERIMENTAL RESULTS

